



Práctica cero: Mecánica pulmonar

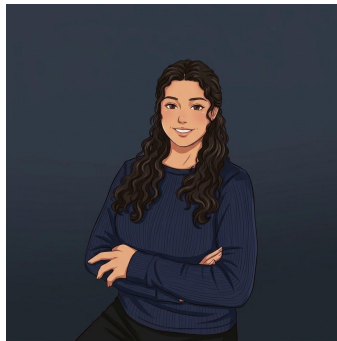
Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

Tecnológico Nacional de México [TecNM - Tijuana], Blvd. Alberto Limón Padilla s/n, C.P. 22454, Tijuana, B.C., México

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Información general



Nombre del alumno: Alexa Montañó Gomez

Número de control: 23212215

Correo institucional: l23212215@tectijuana.edu.mx

Asignatura: **Modelado de Sistemas Fisiológicos**

Docente: **Dr. Paul Antonio Valle Trujillo**; paul.valle@tectijuana.edu.mx

Datos de la simulación

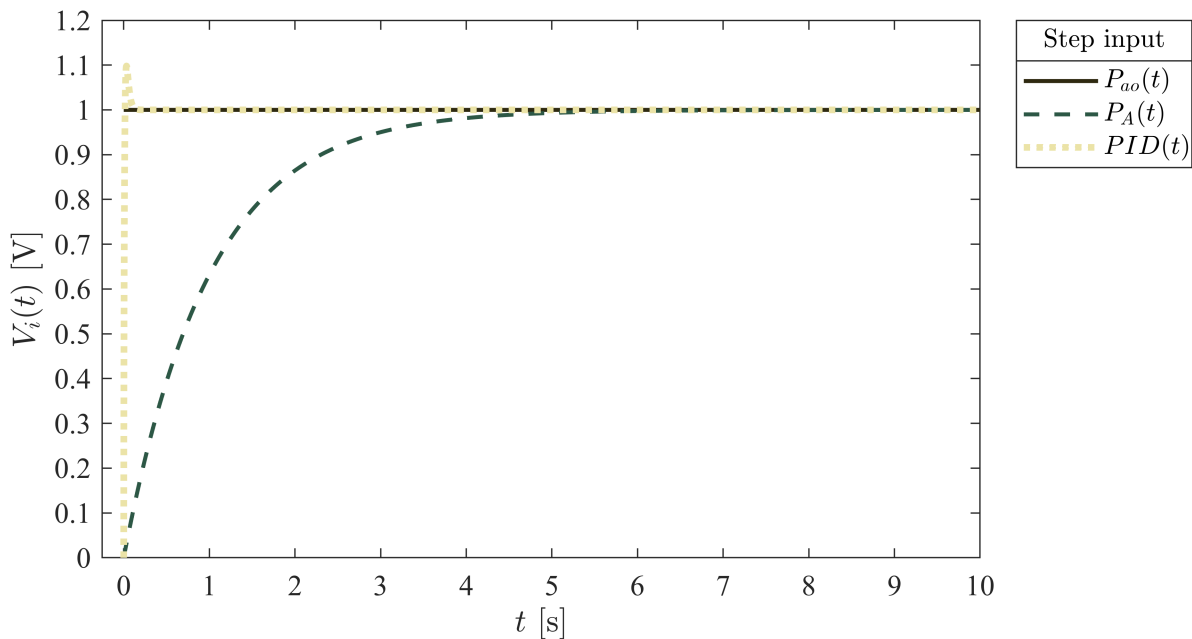
```
clc; clear; close all; warning('off','all')
open_system('Practica_MSf1')
parameters.StopTime = '10';
parameters.Solver = 'ode15s';
parameters.MaxStep = '1e-3';
```

Respuesta al escalón

```
set_param('Practica_MSF1/S1','sw','1')
set_param('Practica_MSF1/Pao(t)','sw','1')
x1 = sim('Practica_MSF1',parameters);
```

Found algebraic loop that contains:
Practica_MSF1/PA(t):EID/Gain2
Practica_MSF1/PA(t):EID/Gain
Practica_MSF1/PA(t):EID/Derivative
Practica_MSF1/PA(t):EID/Add (algebraic variable)

```
plotsignals(x1.t,x1.Pao,x1.PA,x1.PID,'Step')
```

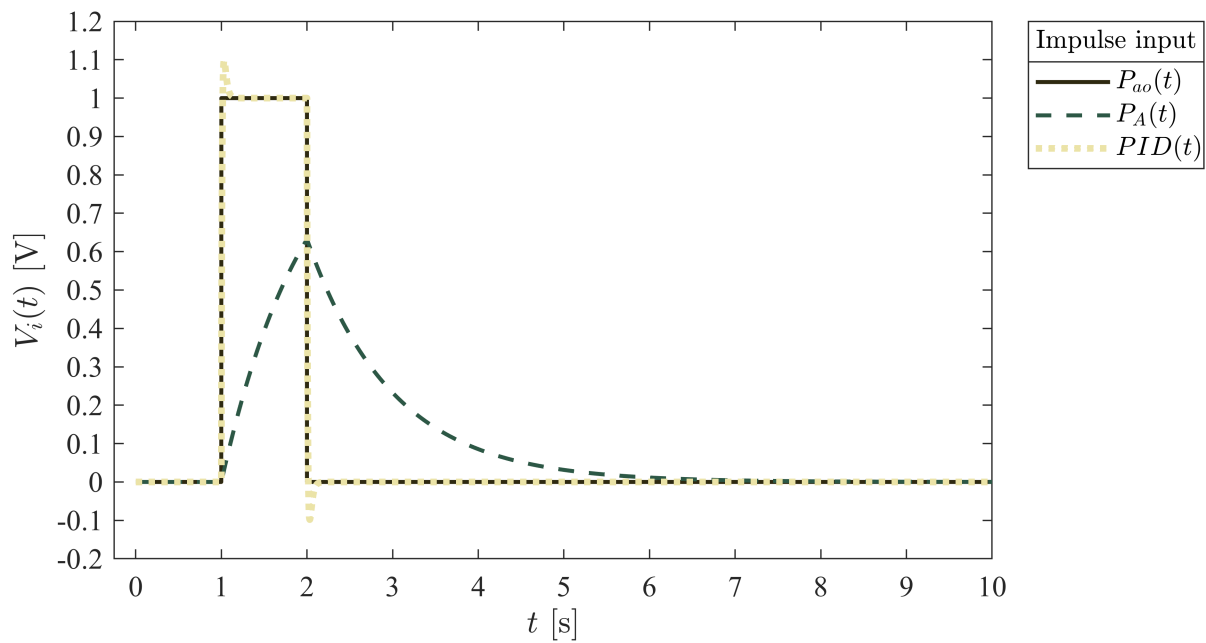


Respuesta al impulso

```
set_param('Practica_MSF1/S1','sw','0')
set_param('Practica_MSF1/Pao(t)','sw','1')
x2 = sim('Practica_MSF1',parameters);
```

Found algebraic loop that contains:
Practica_MSF1/PA(t):EID/Gain2
Practica_MSF1/PA(t):EID/Gain
Practica_MSF1/PA(t):EID/Derivative
Practica_MSF1/PA(t):EID/Add (algebraic variable)

```
plotsignals(x2.t,x2.Pao,x2.PA,x2.PID,'Impulse')
```

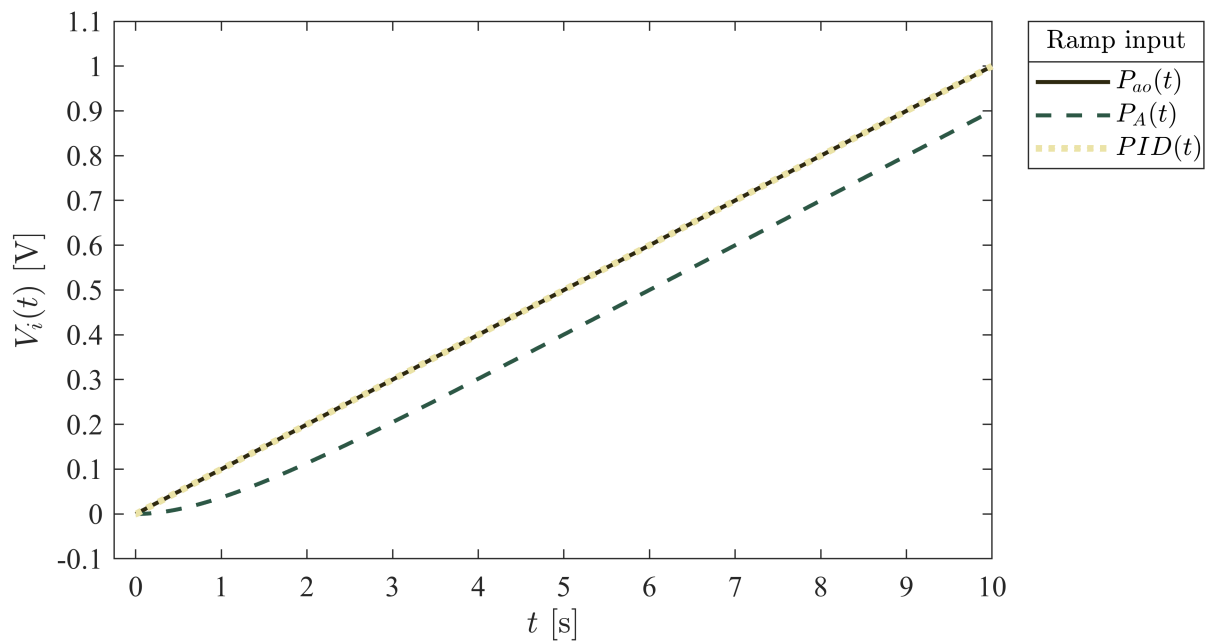


Respuesta a la rampa

```
set_param('Practica_MSF1/S2','sw','1')
set_param('Practica_MSF1/Pao(t)','sw','0')
x3 = sim('Practica_MSF1',parameters);
```

Found algebraic loop that contains:
 Practica_MSF1/PA(t):EID/Gain2
 Practica_MSF1/PA(t):EID/Gain
 Practica_MSF1/PA(t):EID/Derivative
 Practica_MSF1/PA(t):EID/Add (algebraic variable)

```
plotsignals(x3.t,x3.Pao,x3.PA,x3.PID,'Ramp')
```

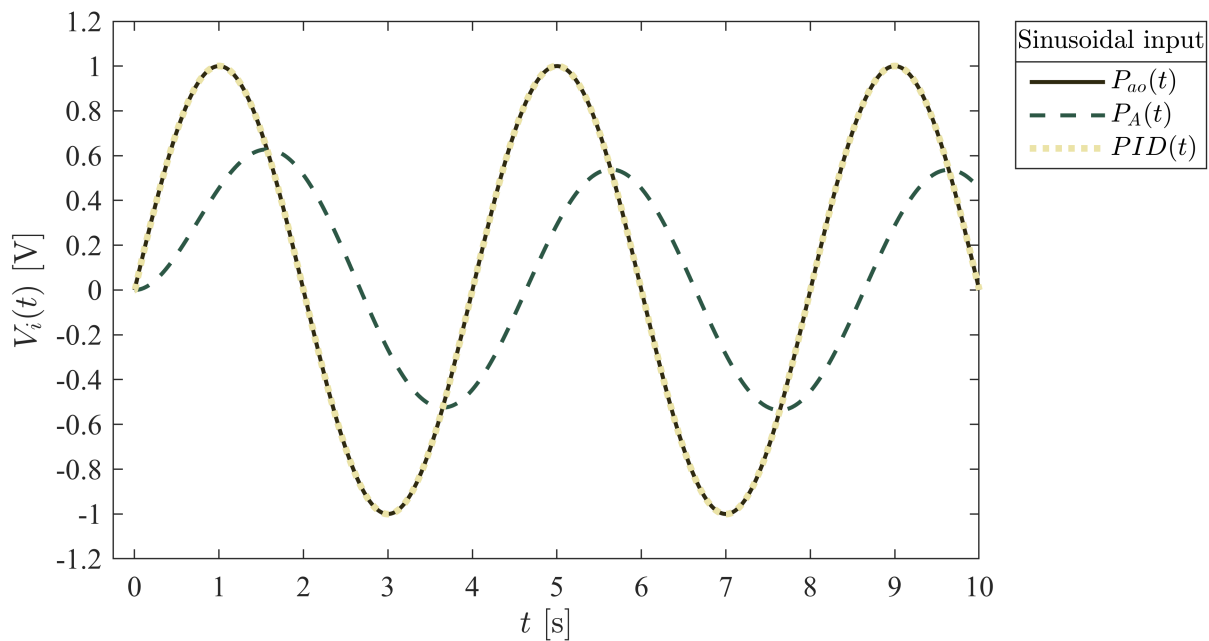


Respuesta a la función sinusoidal

```
set_param('Practica_MSF1/S2','sw','0')
set_param('Practica_MSF1/Pao(t)','sw','0')
x4 = sim('Practica_MSF1',parameters);
```

Found algebraic loop that contains:
 Practica_MSF1/PA(t):EID/Gain2
 Practica_MSF1/PA(t):EID/Gain
 Practica_MSF1/PA(t):EID/Derivative
 Practica_MSF1/PA(t):EID/Add (algebraic variable)

```
plotsignals(x4.t,x4.Pao,x4.PA,x4.PID,'Sinusoidal')
```



Funcion: Respuesta de simulacion

```
function plotsignals(t, Pao, PA, PID, signal)
    set(gcf, 'color', 'w')
    set(gcf, 'units', 'inches', 'position', [1, 1, 7, 3.5])
    set(gca, 'fontname', 'times new roman', 'fontsize', 11)
    hold on; grid off; box on;
    clr1 = [46, 41, 16]/255;
    clr2 = [44, 87, 69]/255;
    clr3 = [235, 227, 167]/255;
    order = [clr1; clr2; clr3]; colororder(order)

    plot(t, squeeze(Pao), '--', 'LineWidth', 1.5, 'Color', clr1)
    plot(t, squeeze(PA), '--', 'LineWidth', 1.5, 'Color', clr2)
    plot(t, squeeze(PID), ':', 'LineWidth', 2.5, 'Color', clr3)

    L = legend('$P_{ao}(t)$', '$P_A(t)$', '$PID(t)$');
    set(L, 'interpreter', 'latex', 'location', 'northeastoutside', 'orientation',
    'vertical')
    title(L, [signal, ' input'])
    xlabel('$t$ [s]', 'interpreter', 'latex')
    ylabel('$V_i(t)$ [V]', 'interpreter', 'latex')
    xlim([-0.25 10]); xticks(0:1:10)

    if strcmp(signal, 'Step')
        ylim([0 1.2]); yticks(0:0.1:1.2)
    elseif strcmp(signal, 'Impulse')
        ylim([-0.2, 1.2]); yticks(-0.2:0.1:1.2)
    elseif strcmp(signal, 'Ramp')
        ylim([-0.1, 1.1]); yticks(-0.1:0.1:1.1)
```

```
elseif strcmp(signal, 'Sinusoidal')
    ylim([-1.2, 1.2]); yticks(-1.2:0.2:1.2)
end

exportgraphics(gcf, [signal, '_matlab.pdf'], 'ContentType', 'vector')
end
```